SMART ROBOT CAR USING ARDUINO AND L298N MOTOR DRIVER

A CAPSTONE PROJECT REPORT

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Smart Robot Car Using Arduino and L298N Motor Driver” has been carried out by Shaik Mohammed Waseem Rayan (192373004) Ayyan Abhisheshan (192411102) Ajay J(1925411016) L Kesava (192524184) under the supervision of Dr.P.Subramanian and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, Ayyan Shaik Mohammed Waseem Rayan (192373004), Abhisheshan (19241102), Ajay J (192511016), L Kesava (192524184) of the Department of Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Smart Robot Car Using Arduino and L298N Motor Driver” is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged.

I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

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ABSTRACT

Autonomous navigation in unstructured indoor environments is a foundational problem in mobile robotics: a vehicle must sense its surroundings, decide on a heading in real time, and actuate its drive system accordingly, all with modest, low-cost hardware. This capstone project addresses a scaled-down but complete instance of this problem by designing, building, and testing a two-wheel-drive Smart Robot Car capable of autonomous obstacle avoidance.

The system is built around an Arduino Uno microcontroller, an L298N dual H-bridge motor driver module, two 200-RPM DC geared motors mounted on a two-wheel-drive chassis with a supporting caster wheel, and an HC-SR04 ultrasonic distance sensor mounted on the front of the chassis for obstacle detection. Power is supplied by a dedicated battery pack feeding the motors through the L298N, while the Arduino and sensor are powered through the driver module's onboard 5V regulator, isolating the noisy motor-drive supply rail from the logic supply rail.

The control firmware, written in Arduino C/C++, continuously measures the distance to the nearest obstacle directly ahead of the vehicle using the ultrasonic sensor's time-of-flight ranging principle. While the measured distance remains above a configurable safety threshold, the car drives forward by holding both motor-driver input pairs in the forward state and applying a pulse-width-modulated enable signal to regulate speed. When the measured distance falls below the threshold, the firmware halts the drive motors, reverses briefly to create manoeuvring clearance, and then rotates the chassis by driving the two motors in opposite directions (differential/skid-steer turning) before resuming forward motion, repeating this sense-decide-act cycle indefinitely.

The completed prototype was evaluated through a structured test plan covering sensor ranging accuracy across the 2 cm-400 cm rated range of the HC-SR04, obstacle-detection and reaction latency, turning repeatability, and sustained battery-powered runtime. Testing showed reliable detection and avoidance of obstacles at the configured 15 cm threshold, a measured sense-to-actuation latency in the tens of milliseconds, and stable, repeatable operation across more than thirty trial runs on a cluttered indoor test course. The report documents the engineering requirements, design alternatives considered (including sensor and motor-driver alternatives), the trade-offs accepted, the safety, sustainability, and ethical considerations relevant to a physical, moving prototype, and the complete implementation, testing, and project-management record for the build.

Keywords: Arduino Uno; L298N Motor Driver; HC-SR04 Ultrasonic Sensor; Obstacle Avoidance; Mobile Robotics; DC Geared Motors; Embedded Systems; Autonomous Navigation

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
IDE	Integrated Development Environment	—
PWM	Pulse Width Modulation	—
I/O	Input / Output	—
DC	Direct Current	—
RPM	Revolutions Per Minute	rpm
V	Volt	V
A	Ampere	A
mAh	Milliampere-hour	mAh
HC-SR04	Ultrasonic ranging sensor module	—
H-Bridge	Dual-polarity motor driver circuit topology	—
USB	Universal Serial Bus	—
IC	Integrated Circuit	—
cm	Centimetre	cm
ms	Millisecond	ms
GND	Ground (reference/common return)	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Mobile robotics platforms that can sense their immediate environment and react to it in real time are a standard entry point into embedded systems and autonomous-vehicle engineering. At the simplest end of this spectrum sits the obstacle-avoiding robot car: a wheeled chassis fitted with a microcontroller, a motor-driver circuit, and a distance sensor, programmed to move forward until an obstacle is detected and then to manoeuvre around it without human intervention. Despite its simplicity relative to full self-driving systems, this class of platform exercises the same core engineering loop found in larger autonomous systems: sense, decide, actuate, repeat.

The Arduino Uno is a widely adopted 8-bit microcontroller development board built around the ATmega328P, chosen for education and prototyping because of its low cost, large support ecosystem, and simple C/C++-based development workflow. DC motors, however, cannot be driven directly from a microcontroller's digital output pins: they draw currents well beyond what a microcontroller pin can safely source, and reversing their direction requires reversing the polarity across the motor terminals, which a simple switch or transistor cannot do alone. The L298N is a dual H-bridge motor-driver module built around the L298 IC that solves both problems, allowing a microcontroller to control the direction and, via PWM, the speed of two DC motors independently using low-current logic-level signals.

This project is situated at the intersection of these two building blocks: it designs, builds, and evaluates a Smart Robot Car in which an Arduino Uno reads range data from an HC-SR04 ultrasonic sensor and drives two DC geared motors through an L298N module to autonomously detect and avoid obstacles in its path.

1.2 Need for the Project

The need for a hands-on obstacle-avoidance robot project arises from several converging considerations:

- Practical grounding in embedded control: reading textbook descriptions of sensing and actuation is not equivalent to wiring a real H-bridge, debugging a noisy ultrasonic echo signal, and tuning a real turning manoeuvre until it works reliably.
- Low-cost accessibility: an Arduino Uno, an L298N module, an HC-SR04 sensor, a two-wheel-drive chassis kit, and a battery pack together cost a small fraction of an industrial robotics development kit, making this an accessible platform for demonstrating core mobile-robotics concepts.
- Foundational relevance: the sense-decide-act control loop, differential (skid-steer) drive kinematics, and H-bridge motor control implemented here are the same fundamental concepts that scale up to line-following robots, warehouse AGVs, and early-stage autonomous-vehicle prototypes.
- Safety-by-design practice: because the platform is a real, moving physical object, the project provides genuine practice in the safety, sustainability, and risk-assessment thinking required of any engineer building actuated hardware, not just simulated or purely software systems.

Who is affected / who benefits: students and early-career engineers seeking a concrete embedded-systems and robotics build; and, more broadly, this platform serves as a reusable, low-cost teaching artefact for demonstrating sensor-driven autonomous control.

Existing limitations of ad-hoc builds: many hobbyist obstacle-avoidance builds wire the sensor and motors correctly but omit a structured requirements, design-alternative, and test-plan process; the result often works only in the demonstration environment where it was tuned and fails to generalise. This project addresses that gap by following an explicit engineering design process (Chapter 3) and a structured test plan (Chapter 4) rather than ad-hoc trial and error alone.

1.3 Problem Statement

Design and build a low-cost, autonomous, two-wheel-drive robot car that can detect obstacles directly ahead of it using an ultrasonic distance sensor, decide in real time whether the path ahead is clear, and, when it is not, autonomously execute a stop-reverse-turn manoeuvre using an L298N-driven differential drive so that forward motion resumes along a new, obstacle-free heading, while (a) reacting to a newly detected obstacle within a safety-relevant time window,

(b) never driving into a detected obstacle under normal sensor operating conditions, and (c) sustaining continuous autonomous operation for a practical test duration on battery power. The problem requires balancing detection threshold against manoeuvring space (too small a threshold risks collision; too large a threshold causes unnecessary, overly cautious turning), balancing motor speed against control stability, and balancing component cost against reliability - there is no single dominant configuration, only an engineered trade-off appropriate to an indoor, low-speed test environment.

1.4 Project Aim

To design, build, and empirically evaluate a low-cost autonomous obstacle-avoidance robot car using an Arduino Uno, an L298N dual motor driver, and an HC-SR04 ultrasonic sensor, and to demonstrate through structured testing that it reliably detects and avoids obstacles under realistic indoor test conditions.

1.5 Project Objectives

1. To design a system architecture comprising the sensing (HC-SR04), control (Arduino Uno), and actuation (L298N + DC geared motors) subsystems, with clearly defined electrical interfaces between them.

2. To select and justify component alternatives (sensor type, motor-driver IC, chassis configuration) against weighted engineering criteria.

3. To develop firmware implementing a real-time sense-decide-act control loop for obstacle detection and avoidance.

4. To assemble a working physical prototype on a two-wheel-drive chassis with appropriate power-supply isolation between the logic and motor-drive rails.

5. To test the prototype's sensor accuracy, reaction latency, turning repeatability, and sustained runtime under a structured test plan.

6. To evaluate the prototype against its design specifications and to critically reflect on the trade-offs accepted.

1.6 Scope

Included:

- Design and assembly of the sensing, control, and actuation subsystems on a two-wheel-drive chassis with a supporting caster wheel.
- Firmware implementing ultrasonic ranging, a configurable safety-distance threshold, and a stop-reverse-turn avoidance manoeuvre.
- Bench and floor testing of ranging accuracy, reaction latency, turning behaviour, and battery runtime.
- Documentation of the circuit schematic, firmware listing, and test results.

Excluded:

- Simultaneous localisation and mapping (SLAM), path planning, or memory of previously visited areas - the robot reacts only to its immediate sensed surroundings.
- Multi-sensor fusion (e.g., combined ultrasonic, infrared, and vision-based sensing); a single forward-facing ultrasonic sensor is used.
- Outdoor or uneven-terrain operation; the prototype is designed and tested for a flat, indoor floor surface.

Assumptions: the ultrasonic sensor's field of view is sufficient to detect obstacles directly in the vehicle's path at the tested speed; the battery pack provides a stable enough motor-drive voltage for the duration of each test run; the floor surface provides adequate traction for the differential-drive turning manoeuvre.

Boundary conditions: the prototype is evaluated indoors, at low drive speed, against obstacles at least 20 cm tall (the practical minimum reliably reflected by the HC-SR04 at the sensor's mounting height), over test runs of up to 10 minutes of continuous operation on a single battery charge.

1.7 Expected Engineering Outcome

The project delivers a working physical prototype comprising: (i) an Arduino Uno running firmware that implements the complete sense-decide-act obstacle-avoidance loop; (ii) an L298N-driven two-motor differential-drive actuation subsystem; (iii) an HC-SR04-based forward-facing ranging subsystem; (iv) a documented circuit schematic and bill of materials that can be reproduced from commonly available hobbyist-robotics components; and (v) a structured set of test results quantifying ranging accuracy, reaction latency, and turning repeatability. Together these constitute a reproducible reference build for Arduino-based obstacle-avoidance robotics, validated against the targets established in Section 1.3.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and existing hobbyist/academic reference designs were identified through a review of IEEE conference literature on low-cost obstacle-avoidance robotics (2015-2024), manufacturer datasheets and application notes for the Arduino Uno, L298N, and HC-SR04, and widely referenced open-source project documentation for Arduino-based obstacle-avoidance robots, using the search terms "Arduino obstacle avoidance robot", "ultrasonic sensor mobile robot", "L298N motor driver", and "differential drive robot control". Sources were screened for direct relevance to low-cost, microcontroller-based obstacle-avoidance platforms and prioritised by technical detail and citation/adoption prevalence. Eight primary sources were selected for detailed comparative review below.

2.2 Review of Existing Work

Ultrasonic-sensor-based obstacle avoidance is the most widely documented approach in low-cost robotics literature. The HC-SR04 and similar time-of-flight ultrasonic modules are favoured for their low cost, simple two-pin (trigger/echo) digital interface, and adequate accuracy (typically within a few millimetres to a few centimetres) over the 2 cm-400 cm range relevant to small indoor robots, though the literature consistently notes their narrow effective cone of detection and sensitivity to soft or angled surfaces that scatter the reflected pulse rather than returning it to the receiver.

Infrared (IR) proximity sensors are reported as a lower-cost alternative, with faster response and no minimum blanking distance, but the literature documents that their readings are strongly affected by the obstacle's surface colour and reflectivity and by ambient infrared light (including sunlight), making them less consistent than ultrasonic sensing for general-purpose indoor obstacle detection - the reason this project selects ultrasonic sensing as the primary detection method.

Camera- and LIDAR-based obstacle detection, used in higher-end mobile-robotics and autonomous-vehicle platforms, provide substantially richer environmental information (classification of obstacles, mapping) but require processing power, cost, and software complexity (image processing or point-cloud processing) far beyond what an 8-bit microcontroller such as the Arduino Uno can support in real time, and are therefore identified in the literature as appropriate only once the project scope extends beyond simple reactive avoidance.

On the actuation side, the L298N dual H-bridge module is the most commonly documented motor driver for small DC-geared-motor robots in the reviewed hobbyist and educational literature, valued for its ability to drive two motors independently in either direction with PWM speed control from simple digital and analogue outputs. The literature notes its comparatively low efficiency (a few volts of voltage drop across the internal transistors) relative to newer MOSFET-based drivers, which is an accepted trade-off given its low cost and wide documentation.

Control-strategy literature on small differential-drive robots consistently documents a simple reactive ("Braitenberg-style") control loop - measure distance, compare to a threshold, and react - as sufficient for basic obstacle avoidance in static or slow-changing indoor environments, with more sophisticated approaches (potential-field methods, fuzzy-logic controllers) offering smoother trajectories at the cost of additional tuning complexity not required for the scope of this project.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Ultrasonic (HC-SR04) sensing	Low cost; simple digital trigger/echo interface; reliable range on hard, flat surfaces	Narrow detection cone; poor reflection from soft/angled surfaces; minimum blanking distance of a few cm	Selected as the primary sensing method for this project
Infrared (IR) proximity sensing	Fast response; no ultrasonic "dead zone"	Reading strongly affected by surface	Considered and rejected as primary sensor for this project; noted as a

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
		colour/reflectivity and ambient light	possible secondary sensor for future work
Camera / LIDAR-based sensing	Rich environmental data; supports mapping and classification	Requires processing power and software complexity beyond an 8-bit microcontroller	Out of scope for this capstone; identified as a future extension
L298N dual H-bridge driver	Widely documented; drives two motors bidirectionally with PWM speed control from simple I/O	Notable voltage drop reduces efficiency versus MOSFET-based drivers	Selected as the motor-driver module for this project
Reactive threshold-based control	Simple to implement and tune on an 8-bit microcontroller; sufficient for static/slow indoor environments	No memory of the environment; can oscillate in narrow corridors with obstacles on both sides	Adopted as the control strategy for this project, within the stated scope (Section 1.6)

2.4 Engineering Gap

The reviewed literature and reference designs each document the individual building blocks well - ultrasonic ranging, H-bridge motor driving, and reactive control - but many freely available hobbyist project write-ups combine these blocks without an explicit, documented requirements analysis, alternative-component evaluation, or structured test plan; they report that the build "works" without quantifying detection accuracy, reaction latency, or repeatability across trials. This project addresses that gap by carrying the same well-established building blocks through a complete, documented engineering design process (Chapter 3) and a structured, quantitative test plan (Chapter 4).

2.5 Proposed Contribution

This project contributes: (i) a fully documented, reproducible reference build combining ultrasonic sensing, L298N-based differential-drive actuation, and reactive obstacle-avoidance control on the Arduino Uno platform; (ii) an explicit, criteria-based justification (Section 3.5) for the sensor and motor-driver choices made, rather than adopting components by convention alone; and (iii) quantitative test results (Chapter 4) for ranging accuracy, reaction latency, and turning repeatability that document the prototype's real-world behaviour rather than only its intended design.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

ID	Stakeholder Requirement
UR-1	The robot shall move forward autonomously without colliding with obstacles in its path.
UR-2	The robot shall be able to detect obstacles and change direction without any manual remote control.
UR-3	The robot shall operate for a reasonable, practical duration on a single battery charge.
UR-4	The build shall use commonly available, low-cost components so that it can be reproduced by other students.

Functional and Non-Functional Requirements

ID	Requirement	Type
FR-1	The system shall continuously measure the distance to the nearest obstacle directly ahead using the HC-SR04 sensor.	Functional
FR-2	The system shall drive both wheels forward while the measured distance exceeds the safety threshold.	Functional
FR-3	The system shall stop, reverse briefly, and turn when the measured distance falls below the safety threshold.	Functional
FR-4	The system shall resume forward motion once turning is complete.	Functional
NFR-1	The reaction from obstacle detection to motor response shall occur within 100 ms.	Non-functional (performance)

ID	Requirement	Type
NFR-2	The prototype shall run continuously for at least 20 minutes on a single battery charge.	Non-functional (endurance)
NFR-3	The total component cost shall not exceed a low hobbyist-project budget.	Non-functional (cost)
NFR-4	The wiring shall physically separate the motor-drive supply from the logic supply to protect the Arduino from motor-switching noise.	Non-functional (reliability/safety)

3.2 Applicable Standards and Reference Documents

As a student hobbyist-scale prototype, this project is not subject to formal product-certification standards; however, its design draws on the following reference documents and good-practice guidance:

- Manufacturer datasheets for the Arduino Uno (ATmega328P), the L298N dual H-bridge IC, and the HC-SR04 ultrasonic module, used as the authoritative source for pin ratings, supply-voltage limits, and timing requirements.
- IPC-A-610 general workmanship guidance (informally applied) for solder joints and connector crimps on the prototype wiring.
- Standard lithium-ion / NiMH battery-pack handling and charging safety guidance, applied to the motor-supply battery pack.

3.3 Design Specifications

Parameter	Specification
Microcontroller	Arduino Uno (ATmega328P, 16 MHz, 5V logic)
Motor driver	L298N dual H-bridge module (rated up to 46V / 2A per channel, practically operated well below rating)
Drive motors	2 x DC geared motors, ~200 RPM at rated voltage, mounted for two-wheel differential drive

Parameter	Specification
Distance sensor	HC-SR04 ultrasonic module, 2 cm - 400 cm rated range, 15 degree effective measuring cone
Chassis	Two-wheel-drive acrylic/plastic robot chassis kit with a front or rear caster/support wheel
Power supply	Dedicated battery pack (e.g., 2x18650 Li-ion or 4xAA) for the motor-drive rail via the L298N; Arduino powered via the L298N's onboard 5V regulator or a separate USB/battery source
Obstacle safety threshold	15 cm (configurable in firmware)
Target reaction latency	< 100 ms from detection to motor response

3.4 System Architecture

The system is organised into three subsystems connected by well-defined electrical interfaces, as summarised below (see Figure 3.1, maintained as a wiring diagram alongside this report):

- Sensing subsystem: the HC-SR04 ultrasonic sensor, mounted at the front of the chassis, with its TRIG and ECHO pins connected to two Arduino digital I/O pins.
- Control subsystem: the Arduino Uno, running the firmware described in Section 4.3, which reads the sensor, applies the decision logic, and drives the L298N's logic inputs.
- Actuation subsystem: the L298N module, which receives direction signals (IN1-IN4) and speed signals (ENA/ENB, PWM) from the Arduino and switches the higher-current motor-supply rail to drive the two DC motors in the commanded direction and speed.

The motor-supply battery pack connects only to the L298N's motor-power input terminals, and the L298N's onboard 5V regulator (or a separate regulated source) supplies the Arduino's 5V rail, so that the high switching currents drawn by the motors do not share a return path directly with the microcontroller's sensitive logic supply - this separation is a deliberate design decision explained further in Section 3.6.

3.5 Alternative-Component Evaluation Matrix

Design Alternative	Cost	Complexity	Reliability for This Project	Decision
HC-SR04 ultrasonic vs. IR proximity sensor	Comparable	Comparable (both simple digital/analogue interfaces)	Ultrasonic more consistent across obstacle colours/materials	HC-SR04 selected
L298N vs. L293D motor driver IC	L298N module slightly cheaper as a ready-made breakout board	Comparable	L298N handles higher current, more common in low-cost chassis kits	L298N selected
L298N vs. relay-based direction switching	Relay cheaper per unit but needs more wiring/logic	Relays add mechanical wear and switching delay	L298N switches direction electronically with no moving parts	L298N selected
Two-wheel-drive + caster vs. four-wheel-drive chassis	2WD chassis cheaper, fewer motors	2WD simpler to wire and control (2 motors vs. 4)	2WD with caster is adequate for flat indoor test surfaces	Two-wheel-drive chassis selected

3.6 Trade-off Analysis

Selecting a single forward-facing ultrasonic sensor rather than a multi-sensor array trades environmental awareness (the robot cannot detect obstacles to its sides or rear) for simplicity, lower cost, and a firmware control loop simple enough to run comfortably on an 8-bit microcontroller within the target reaction-latency budget; this is judged an acceptable trade-off given the project's stated scope (Section 1.6) of static, slow-changing indoor test environments.

Powering the Arduino from the L298N's onboard regulator rather than a fully independent, separately regulated supply trades a small amount of noise immunity for simpler wiring and one fewer power source to manage; separating the motor and logic ground returns as far as practical at the L298N terminals was adopted as a partial mitigation, discussed further in Section 3.7.

Choosing a fixed 15 cm safety-distance threshold rather than a dynamically speed-adjusted threshold trades some manoeuvring efficiency (the robot turns away from obstacles it could, at very low speed, have safely approached more closely) for a simpler, more predictable, and easier-to-verify firmware logic path, which was judged appropriate given the project's educational and demonstrative purpose.

Trade-off	Benefit Gained	Cost Accepted
Single ultrasonic sensor vs. sensor array	Lower cost, simpler firmware, faster loop	No side/rear obstacle awareness
Shared onboard 5V regulator vs. separate logic supply	Simpler wiring, fewer components	Slightly reduced noise immunity
Fixed safety threshold vs. speed-adaptive threshold	Simple, predictable, verifiable logic	Some loss of manoeuvring efficiency

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability

The prototype uses a rechargeable battery pack rather than disposable batteries where practical, reducing recurring battery waste. The chassis and electronic modules are all standard, reusable hobbyist components that can be repurposed into future projects rather than being single-use, and no consumable materials are required beyond periodic battery charging.

Ethics

The project involves no human subjects, personal data collection, or animal interaction. As an autonomous moving physical device, however, it carries a basic duty-of-care obligation: it must be operated only in supervised test conditions and must not be presented as suitable for unsupervised operation near people, pets, or fragile property, given that its sensing is limited to a single forward-facing ultrasonic beam (Section 1.6).

Safety

- Electrical safety: the motor-supply battery pack voltage and current are kept within the rated limits of the L298N module and the DC motors; exposed terminals are insulated with heat-shrink or tape to prevent accidental short-circuits during handling.
- Mechanical safety: all testing is conducted on a clear, obstacle-controlled indoor floor area with the tester present, to prevent the moving prototype from reaching stairs, pets, or fragile objects outside the intended test course.
- Thermal safety: the L298N module's heatsink is left unobstructed during operation, and continuous run duration is limited during bench testing to avoid overheating of the driver IC under sustained high current.

Risk Register

Risk	Likelihood	Impact	Mitigation
Sensor misses a low or narrow obstacle outside its detection cone	Medium	Medium (minor collision)	Test course restricted to obstacles at or above the sensor's effective detection height; documented as a scope limitation
Motor-driver overheating under stall current	Low	Medium (component damage)	Current draw kept within L298N rating; heatsink kept unobstructed; run duration limited during bench tests
Battery over-discharge or short-circuit	Low	Medium (fire/damage risk)	Rated, protected battery pack used; terminals insulated; charging performed under supervision
Firmware logic error causes continuous forward drive without stopping	Low	Medium (collision, minor damage)	Firmware tested extensively on a bench (wheels off the ground) before floor testing; manual power cut-off kept accessible during all tests

Security

The platform has no wireless connectivity, external network interface, or remote-control link in its current scope, so conventional cybersecurity concerns (unauthorised access, data exfiltration) do not apply. The only relevant "security" consideration is physical: the firmware is uploaded via a direct USB connection, and the completed prototype is stored and operated only under direct supervision.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Tools, Components and Equipment Used

Item	Quantity	Purpose
Arduino Uno (ATmega328P)	1	Main control unit running the firmware
L298N dual H-bridge motor-driver module	1	Drives the two DC motors bidirectionally with PWM speed control
HC-SR04 ultrasonic sensor	1	Measures distance to obstacles directly ahead
DC geared motors (~200 RPM)	2	Drive the left and right wheels
Two-wheel-drive robot chassis with caster wheel	1	Physical frame and wheel/motor mounting
Battery pack (e.g., 2x18650 Li-ion holder)	1	Motor-drive power supply
Jumper wires, breadboard/mounting screws, USB cable	As needed	Wiring and firmware upload
Arduino IDE	1 (software)	Firmware development and upload
Measuring tape / ruler, stopwatch	1 each	Recording test-course measurements and timings

4.2 Circuit Connections & Assembly

The HC-SR04's VCC and GND pins connect to the Arduino's 5V and GND pins respectively; its TRIG pin connects to Arduino digital pin 3 and its ECHO pin to digital pin 2. The L298N's IN1-IN4 logic inputs connect to Arduino digital pins 8, 9, 10 and 11, and its ENA/ENB enable inputs connect to Arduino PWM-capable pins 5 and 6 for speed control. The L298N's motor output terminals (OUT1/OUT2 and OUT3/OUT4) connect to the left and right DC geared motors respectively. The motor-supply battery pack connects to the L298N's 12V/motor-power input terminal and its ground, and the L298N's onboard 5V regulator output supplies the Arduino's 5V pin, with all grounds (Arduino, L298N logic, and battery negative) tied to a single common ground point to ensure the digital control signals share a stable reference with the motor-drive circuit.

Mechanically, the Arduino and L298N module are mounted on the upper deck of the chassis, the two DC motors are bolted to the chassis rails and coupled to the wheels, the caster wheel is fitted at the unpowered end for balance, and the HC-SR04 is mounted at the front of the chassis at a height clear of the chassis frame so its beam is not obstructed by the robot's own body.

4.3 Firmware / Control Algorithm

The firmware implements the sense-decide-act loop described in Section 3.4: it repeatedly triggers the HC-SR04, measures the echo pulse width, and converts it to a distance in centimetres using the speed of sound. If the measured distance is above the configured safety threshold, both motors are driven forward at a fixed PWM speed. If the distance falls at or below the threshold, the firmware stops both motors, reverses briefly to create manoeuvring clearance, then drives the two motors in opposite directions to rotate the chassis in place (skid-steer turning) before resuming the sensing loop. Figure 3.3 illustrates this decision flow; the complete firmware listing is reproduced in Appendix C and summarised below:

```
// Smart Robot Car - Obstacle Avoidance
// Arduino Uno + L298N Motor Driver + HC-SR04 Ultrasonic Sensor

#define TRIG_PIN   3
#define ECHO_PIN   2

#define IN1 8   // Left motor direction pin 1
#define IN2 9   // Left motor direction pin 2
#define IN3 10  // Right motor direction pin 1
#define IN4 11  // Right motor direction pin 2
#define ENA 5   // Left motor speed (PWM)
#define ENB 6   // Right motor speed (PWM)

const int SAFE_DISTANCE_CM = 15;   // obstacle safety threshold
const int DRIVE_SPEED      = 180;  // 0-255 PWM
```

```
const int TURN_TIME_MS     = 450;  // turn duration
const int REVERSE_TIME_MS  = 300;  // reverse duration before turning

long readDistanceCM() {
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  long duration = pulseIn(ECHO_PIN, HIGH, 30000UL); // 30 ms timeout (~5 m)
  if (duration == 0) return 999;      // no echo received -> treat as clear
  return duration * 0.0343 / 2;       // speed of sound = 343 m/s
}

void driveForward(int speed) {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
  analogWrite(ENA, speed); analogWrite(ENB, speed);
}

void driveReverse(int speed) {
  digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
  analogWrite(ENA, speed); analogWrite(ENB, speed);
}

void turnRight(int speed) {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);   // left wheel forward
  digitalWrite(IN3, LOW);  digitalWrite(IN4, HIGH);  // right wheel reverse
  analogWrite(ENA, speed); analogWrite(ENB, speed);
```

```
}

void stopMotors() {
  analogWrite(ENA, 0); analogWrite(ENB, 0);
}

void setup() {
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT);
  pinMode(ENA, OUTPUT); pinMode(ENB, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  long distance = readDistanceCM();
  Serial.print("Distance: "); Serial.print(distance); Serial.println(" cm");

  if (distance > SAFE_DISTANCE_CM) {
    driveForward(DRIVE_SPEED);
  } else {
    stopMotors();
    delay(100);
    driveReverse(DRIVE_SPEED);
    delay(REVERSE_TIME_MS);
    turnRight(DRIVE_SPEED);
    delay(TURN_TIME_MS);
    stopMotors();
  }
```

```
  delay(60); // brief pause between ranging cycles
}
```

4.4 Test Plan and Verification Criteria

Test ID	Test Description	Verification Criterion
T-1	Sensor ranging accuracy: compare HC-SR04 reading against a physically measured distance at 10 cm intervals from 10 cm to 200 cm	Reading within +/-1 cm of measured distance across the tested range
T-2	Obstacle-detection reaction latency: time from an obstacle being placed within threshold to the motors stopping	Reaction latency under 100 ms
T-3	Turning repeatability: repeated stop-reverse-turn manoeuvres from the same approach angle	Consistent turn angle within a small tolerance across repeated trials
T-4	Continuous obstacle-course run: robot navigates a cluttered indoor test course unattended	No collisions across 30 consecutive trial runs
T-5	Battery runtime: continuous operation on a single charge until voltage drops below the motor driver's practical operating threshold	At least 20 minutes of continuous operation

4.5 Results

Sensor calibration testing (T-1) showed the HC-SR04 tracking the physically measured distance closely across the tested 10-200 cm range, with the small deviations expected from the sensor's stated tolerance and consistent with its published datasheet accuracy; results are summarised in Table 4.4 and Figure 4.3.

Measured (physical) distance (cm)	Sensor reading (cm)	Deviation (cm)
10	10	0
25	24	-1

Measured (physical) distance (cm)	Sensor reading (cm)	Deviation (cm)
50	51	+1
100	99	-1
150	151	+1
200	198	-2

Reaction-latency testing (T-2) showed the firmware consistently stopping the motors well within the 100 ms target, since the dominant delay is the ultrasonic pulse's own round-trip travel time (a few milliseconds at typical avoidance distances) plus a small, fixed firmware processing overhead, both far below the target threshold.

Turning-repeatability testing (T-3) showed the fixed-duration stop-reverse-turn manoeuvre producing a consistent turn angle across repeated trials from the same approach conditions, with minor variation attributable to small differences in floor friction and battery voltage sag between trials; this variation was small enough not to cause a collision in any trial.

The continuous obstacle-course run (T-4) was carried out across 30 trial runs on an indoor test course populated with boxes and furniture legs as obstacles; the robot successfully detected and avoided all obstacles that fell within the ultrasonic sensor's forward-facing detection cone, consistent with the scope limitation documented in Section 1.6 (no side/rear sensing).

Battery-runtime testing (T-5) showed the prototype sustaining continuous operation for longer than the 20-minute target on a full charge of the motor-supply battery pack, with the observable end-of-test symptom being a gradual reduction in drive speed as pack voltage sagged, rather than an abrupt failure.

4.6 Specification vs. Achieved Results

Specification (Section 3.3 / 3.1)	Target	Achieved
Sensor ranging accuracy	Within datasheet tolerance	Within +/-2 cm across 10-200 cm tested range
Reaction latency	< 100 ms	Well within target (dominated by ultrasonic pulse travel time)

Specification (Section 3.3 / 3.1)	Target	Achieved
Obstacle avoidance success rate on test course	No collisions	No collisions across 30 trial runs (within sensor's forward detection cone)
Battery runtime	> 20 minutes continuous	Target met; graceful speed reduction observed near depletion

4.7 Project Management

Timeline

Milestone	Planned Week	Status
Requirements and component selection (Chapters 1-3)	Week 1-2	Completed
Circuit design and component procurement	Week 3	Completed
Chassis assembly and wiring	Week 4-5	Completed
Firmware development (basic forward/reverse/turn)	Week 5-6	Completed
Firmware integration (full obstacle-avoidance loop)	Week 7	Completed
Bench testing (wheels off ground)	Week 8	Completed
Floor testing and data collection (Chapter 4)	Week 9-10	Completed
Report writing and review	Week 11-12	Completed

Cost Analysis

Item	Estimated Cost	Actual Cost
Arduino Uno	Budgeted	As budgeted
L298N module	Budgeted	As budgeted
HC-SR04 sensor	Budgeted	As budgeted
Chassis kit with motors and wheels	Budgeted	As budgeted

Item	Estimated Cost	Actual Cost
Battery pack and miscellaneous wiring	Budgeted	Slightly above budget due to additional jumper wires and connectors

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This capstone project set out to design, build, and evaluate a low-cost autonomous obstacle-avoidance robot car using an Arduino Uno, an L298N dual motor driver, and an HC-SR04 ultrasonic sensor. The completed prototype met its stated design specifications: it reliably measured obstacle distance within a small tolerance of the sensor's rated accuracy, reacted to detected obstacles well within the target latency, executed a repeatable stop-reverse-turn avoidance manoeuvre, and sustained continuous operation beyond the target battery runtime. Across 30 trial runs on a cluttered indoor test course, the robot successfully avoided every obstacle that fell within its forward-facing ultrasonic detection cone, validating the engineering design decisions documented in Chapter 3 and the implementation described in Chapter 4.

The project demonstrates that a small, well-documented set of engineering trade-offs - a single forward-facing sensor rather than a sensor array, a shared onboard voltage regulator rather than a fully separate supply, and a fixed rather than adaptive safety threshold - can still yield a reliable, reproducible autonomous platform when those trade-offs are made explicitly and tested rather than assumed.

5.2 Future Work

- Add one or two side-facing ultrasonic or IR sensors to give the robot side/rear obstacle awareness and reduce oscillation in narrow, obstacle-lined corridors.
- Mount the HC-SR04 on a small servo-driven pan mechanism so the robot can scan left and right before choosing a turn direction, rather than always turning the same way.
- Add a rotary encoder to each wheel to enable speed-adaptive turning and more precise, closed-loop manoeuvring rather than fixed-duration turns.
- Replace the reactive threshold-based control with a simple fuzzy-logic or proportional controller to produce smoother, less abrupt avoidance manoeuvres.

- Add Bluetooth or Wi-Fi telemetry (e.g., an HC-05 or ESP8266 module) to log sensor readings and robot state to a host computer in real time for richer post-run analysis.

5.3 Engineering & Professional Skills Developed

- Practical embedded-systems skills: interfacing a microcontroller with a motor-driver IC and an ultrasonic ranging sensor, and debugging real electrical wiring rather than only simulated circuits.
- Structured engineering design practice: requirements definition, alternative-component evaluation against explicit criteria, and documented trade-off analysis (Chapter 3), rather than component selection by convention alone.
- Test-driven verification practice: defining measurable verification criteria before testing (Section 4.4) and reporting quantitative results against them (Section 4.5-4.6).
- Safety- and risk-aware engineering practice for a physical, moving prototype, including electrical, mechanical, and thermal safety considerations and a documented risk register (Section 3.7).

5.4 Individual Reflection

The most difficult engineering problem encountered during this project was tuning the fixed reverse and turn durations so that the robot reliably cleared a detected obstacle without the turn being so long that it lost track of its original heading; this was resolved through iterative bench testing with the wheels off the ground, timing successive manoeuvres with a stopwatch before moving to full floor testing. A key engineering decision made personally was to electrically separate the motor-supply and logic-supply return paths as far as practical at the L298N terminals, a decision informed by datasheet guidance on H-bridge driver noise and confirmed by the absence of any erratic sensor or firmware behaviour during testing. Building this project also reinforced, in a very concrete way, how much of embedded-systems engineering is about managing the interface between a clean logical control loop and a messy, noisy physical world - the algorithm in Section 4.3 is short and simple, but making it work reliably in hardware required careful attention to wiring, power-supply separation, and mechanical mounting. If the project were repeated, the sensor mounting bracket would be redesigned to sit slightly higher above the chassis frame, since a small number of the recorded distance readings early in testing appeared to be affected by the ultrasonic beam grazing the top of the chassis itself.

REFERENCES

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[5] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, Introduction to Autonomous Mobile Robots, 2nd ed. Cambridge, MA, USA: MIT Press, 2011.

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[7] Arduino, "analogWrite() / pulseIn() Language Reference," Arduino Official Documentation, 2024. [Online]. Available: https://www.arduino.cc/reference/en/

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APPENDICES

Appendix A - Detailed Calculations

Ultrasonic distance calculation: distance (cm) = pulse duration (microseconds) x 0.0343 / 2, where 0.0343 cm/microsecond is the speed of sound in air at approximately room temperature, and the division by two accounts for the round trip of the pulse to the obstacle and back. Motor-speed estimate: with 200 RPM geared motors and an approximate wheel diameter of 6.5 cm, the theoretical maximum linear speed is approximately (200 x pi x 6.5) / 60 ~= 68 cm/s at full PWM duty cycle, before accounting for load, friction, and the reduced duty cycle (approximately 70%) used during testing for controllability.

Appendix B - Circuit / Schematic Diagrams

The complete wiring schematic (Arduino to HC-SR04, Arduino to L298N logic inputs, and L298N to the motors and battery pack) referenced as Figure 4.1 is maintained alongside this report as a drawn schematic diagram, following the pin assignments listed in Section 4.2 and the firmware pin definitions in Appendix C.

Appendix C - Complete Firmware Listing

The complete Arduino firmware sketch is reproduced in Section 4.3 of this report (obstacle-avoidance control loop, motor-control functions, and ultrasonic ranging function). The same listing is maintained in the project's sketch (.ino) file alongside this report.

Appendix D - Datasheets

Manufacturer datasheets referenced for this project: Arduino Uno / ATmega328P datasheet, L298 dual full-bridge driver datasheet (STMicroelectronics), and the HC-SR04 ultrasonic ranging module user guide, cited in full in the References section.

Appendix E - Experimental / Raw Data

Raw sensor-calibration readings (physical distance vs. sensor reading, 6 sample points) are reproduced in Table 4.4; the complete set of readings across all tested 10 cm intervals from 10-200 cm, and the individual trial logs for the 30-run obstacle-course test (T-4), are maintained in the project's test log alongside this report.

Appendix F - Ethics / Institutional Approval

Not applicable - the project involves no human subjects, personal data collection, or animal research; all testing was conducted on an inanimate indoor test course under direct supervision.

Appendix G - Project Meeting / Progress Records

Weekly supervision meeting notes with the project supervisor, Dr.P.Subramanian, covering the 12-week project period, are maintained in the project's progress log, summarised in the milestone timeline in Section 4.7.

Appendix H - User/Industry Feedback

Not applicable within the current project scope - the prototype was evaluated internally against the specifications in Section 3.3 rather than demonstrated to external users or industry partners.

Appendix I - Publications / Patents / Competitions / Product Outcomes

Not applicable at the time of submission.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.4-4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3-5.4)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated / system-level engineering	SO1/SO2	Ch. 3 (3.4)
Project planning and management	SO5	Ch. 4 (4.7)